VISHNU SURESHKUMAR

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PROFESSIONAL SUMMARY

Cyber security engineer pursuing my MSc in Cybersecurity and Artificial Intelligence at the University of Sheffield with experience in AI security research, threat modelling, penetration testing, vulnerability assessment, secure software development, and adversarial analysis. Researched attacks against AI-enabled and autonomous systems, including data poisoning, neural-network backdoors, bit-flip attacks, sensor spoofing, false-data injection, and model integrity risks. Strong Python programming and technical documentation skills, with additional experience in network traffic analysis, security testing, cryptography, and automated testing. Interested in analysing emerging AI threats, evaluating system misuse cases, and improving the safety, reliability, and security of intelligent systems.

EDUCATION

MSc Cybersecurity and Artificial Intelligence — **September 2025 – September 2026**
University of Sheffield — *Sheffield, United Kingdom*

Expected Grade: 2:1

Bachelor of Technology in Cybersecurity and Computer Science — **November 2021 – May 2025**
Rajadhani Institute of Engineering and Technology — *Kerala, India*

CGPA: 7.28/10

SKILLS

AI Security: Adversarial machine learning, threat modelling, AI risk analysis, model integrity, data poisoning, neural-network backdoors, model extraction concepts, prompt injection concepts, jailbreak concepts, AI misuse analysis, LLM red teaming.

Security Analysis: Vulnerability assessment, penetration testing, attack-surface analysis, security research, incident investigation, network traffic analysis, risk assessment, remediation reporting

Security Frameworks and Concepts: OWASP Top 10, MITRE ATT&CK concepts, secure design principles, trust boundaries, authentication, access control, cryptographic integrity, defence in depth

Programming and Machine Learning: Python, Java, C, SQL, PyTorch, PySpark

Security Tools: Burp Suite, Wireshark, Nmap, Autopsy

Professional Skills: Technical writing, structured analysis, research, independent working, problem solving, communication, documentation, and presenting findings

PROJECTS

Security Analysis of AI-Enabled Autonomous Systems

- Conducted a structured security analysis of AI-enabled autonomous and embedded systems across perception, computation, communication, control-loop, and actuation layers.
- Investigated adversarial threats including data poisoning, neural-network backdoors, model-reuse attacks, bit-flip attacks, GPS spoofing, LiDAR spoofing, false-data injection, and communication-layer attacks.
- Developed a security taxonomy that classified attack mechanisms, affected system layers, safety impact, real-world relevance, and AI-specific characteristics.
- Produced technical documentation, comparison tables, threat models, research summaries, and presentation materials for academic and technical audiences.

Zero-Knowledge Verification of Neural Networks

- Analysed VerfCNN and ZK-Torch as approaches for verifying neural network computations using zero-knowledge proof techniques.
- Investigated how model predictions can be verified while limiting exposure of sensitive model parameters, inputs, or internal computations.
- Compared evaluation criteria including prover time, proof size, verification methodology, supported model operations, and computational overhead by building a common benchmark CNN model to compare them.
- Documented architectural differences, experimental outputs, security trade-offs, and limitations through comparative technical analysis.

EntropyGuard – Behaviour-based Ransomware Detection Prototype

- Developed EntropyGuard, a Python-based behaviour-driven ransomware detection prototype using Shannon entropy

analysis and SHA-256 file integrity verification.
- Designed a SQLite baseline database to track file hashes, metadata, and entropy values for detecting suspicious changes over time.
- Applied digital forensics concepts by collecting file metadata, timestamps, and integrity evidence to support post-incident analysis.
- Built a modular detection framework with extensible components for behavioral analysis, file system monitoring, and future real-time threat detection.

Automotive CAN Bus Authentication System
- Designed a lightweight authentication mechanism to control access to an automotive Controller Area Network through an OBD-II interface.
- Combined vehicle identifiers, password-based authentication, temporary codes, and SHA-256 hashing to generate secure access credentials.
- Simulated legitimate and malicious CAN traffic to evaluate spoofing, replay, unauthorised access, and message-injection scenarios.
- Analysed system trust assumptions, attack surfaces, authentication weaknesses, and message-integrity risks.
- Documented the design, security rationale, testing methodology, observed limitations, and recommended improvements.

EXPERIENCE

Cybersecurity Analyst Intern **January 2025 – February 2025**
TechbyHeart *Thiruvananthapuram, India*

- Supported vulnerability assessments and penetration-testing activities to identify security weaknesses in applications, systems, and network environments.
- Used Burp Suite and Wireshark to inspect application traffic, analyse network behaviour, and investigate potentially suspicious activity.
- Assisted with incident investigation, security event analysis, risk assessment, and malware-analysis fundamentals.
- Researched vulnerabilities, attack techniques, and security controls to support assessments and internal knowledge development.
- Documented findings, potential impact, evidence, and recommended remediation actions in a clear and structured format.
- Worked with senior security professionals and incorporated feedback into technical analysis and reporting.

Engineering Tutor **January 2025 – August 2025**
IPS Engineering Academy *Thiruvananthapuram, India*

- Delivered technical lessons in computer networks, C programming, object-oriented programming with Java, and software-system management.
- Explained complex technical concepts clearly to students with different levels of experience.
- Prepared structured learning materials, organised problem-solving sessions, and provided detailed technical feedback.
- Managed multiple learners and topics independently, demonstrating organisation, analytical thinking, communication, and time management.

CERTIFICATIONS AND LEADERSHIP

Certification: Certified Ethical Hacker, EC-Council – Score: 123/125

Leadership: Chairman, Computer Society of India, RIET Chapter, 2024–2025; Secretary, 2022–2024

Technical Activities: Organised an introductory reconnaissance workshop, an Android development workshop, and inter-college technical debate events.

Research Communication: Presented technical work on AI-enabled embedded-system security, attack taxonomies, runtime defence mechanisms, and cross-layer threat analysis.

ADDITIONAL INFORMATION

Languages: English – Fluent; Malayalam – Native

Working Preference: Comfortable working independently, remotely, asynchronously, and on task-based technical assignments.